

A σ -Essential Contextual State in ZFC: The Product Ulam Carrier

Working draft σ unreferreed

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A σ -Essential Contextual State in ZFC: The Product Ulam Carrier

Status. Candidate theorem, unreferreed. Prepared by Claude (Anthropic) in working sessions with P. S. Mahon, as a companion to *A σ -Essential Contextual State: Localizing an Open Problem on Concrete Orthomodular Lattices* (cited as **[M]**). Every proof below is written to be attacked; §8 lists the load-bearing joints in descending order of residual suspicion and gives Lean encoding suggestions. The theorem resolves the **amended** form of [M, Question 5.1] (σ -class carrier, finite coherence read globally, irreducibility read modulo the countable ideal); the literal OML form remains open (§9).

0. Notation

Symbol	Meaning
M	ω_1 (any set of cardinality $\aleph_1$ with a fixed well-order of type ω_1)
F	the four-element fiber set $\{1, 2, 3, 4\}$
Ω	$M \times F$

Symbol	Meaning
E_f	fiber trace: $E_f = \{x \in M : (x, f) \in E\}$, for $E \subseteq \Omega, f \in F$
$X \approx Y$	$X \Delta Y$ is countable ($X, Y \subseteq M$)
$\mathbb{F}_2^4, E_4$	binary quadruples; $E_4 =$ even-weight code $= \{0000, 1111\} \cup \{\text{six weight-2 vectors}\}$
N	normalized cosets: $N = \{\kappa \in E_4 : \kappa_4 = 0\} = \{0000, 1100, 1010, 0110\}$
$\xi^{(0)}, \xi^{(1)}$	$\xi^{(0)} = \xi, \xi^{(1)} = M \setminus \xi$, for $\xi \subseteq M$
A, B, C	cores: $A = M \times \{1, 2\}, B = M \times \{1, 3\}, C = M \times \{2, 3\}$

Coordinate order for trace quadruples is $(1, 2, 3, 4)$. Thus A, B, C have trace patterns 1100, 1010, 0110 respectively, and $A \cap B = M \times \{1\}$, $A \cap C = M \times \{2\}$, $B \cap C = M \times \{3\}$, $A \cap B \cap C = \emptyset$, $(A \cup B \cup C)' = M \times \{4\}$. The three “solo” regions $A \cap B' \cap C'$, $A' \cap B \cap C'$, $A' \cap B' \cap C$ are empty; the four fibers are exactly the four inhabited Boolean regions of $\{A, B, C\}$.

1. Definitions (amended forms)

Definition 1.1 (σ -class). A σ -class (concrete σ -complete orthomodular poset) on Ω is a family $L \subseteq \mathcal{P}(\Omega)$ with $\emptyset, \Omega \in L$, closed under complement, and closed under countable pairwise-disjoint unions. Order is inclusion; $E^\perp = \Omega \setminus E$.

Definition 1.2 (state). A two-valued state on a $\perp$ -closed subfamily $B \subseteq \mathcal{P}(\Omega)$ is $s : B \rightarrow \{0, 1\}$ with $s(\Omega) = 1$ such that for every **finite** pairwise-disjoint family $\{A_i\} \subseteq B$ with $\bigcup_i A_i \in B$: $s(\bigcup_i A_i) = \sum s(A_i)$. It is σ -additive if the same holds for countable such families. Point evaluations δ_ω are *Dirac*; all others *non-Dirac*.

Definition 1.3 (finite coherence — the amendment). Let $B \subseteq L$ be a finite $\perp$ -closed sub-orthoposet and s_0 a two-valued state on B . Say s_0 is *finitely coherent on L* if s_0 extends to a (finitely additive) two-valued state on all of L .

Definition 1.4 (σ -essential contextual state, amended). s_0 is a σ -essential contextual state on L if s_0 is finitely coherent on L but extends to **no** σ -additive two-valued state on L .

Remark 1.5 (why the amendment is forced). Under the literal reading of [M, Def. 1.1] (“state on finite B ”, no global coherence clause), [M, Prop. 2.1] is false: on $\Omega_7 = \{\pm\}^3 \setminus \{(+, +, +)\}$ with $L = \mathcal{P}(\Omega_7)$ and B the $\perp$ -closure of the three coordinate events, the pattern $s_0 \equiv 1$ on the three events is a two-valued state on B (only complement pairs constrain it), yet $K(s_0) = \emptyset$ and every two-valued state on a finite power set is Dirac. Definition 1.3 restores the baseline:

Proposition 1.6 (Boolean baseline, amended). If L is a Boolean σ -algebra of subsets of Ω and s_0 (on finite $\perp$ -closed $B \subseteq L$) is finitely coherent on L , then s_0 extends to a Dirac (hence σ -additive) state.

Proof. Let $\mu \supseteq s_0$ be a global finitely additive two-valued state on L and $B_0 \subseteq L$ the finite Boolean subalgebra generated by B . The atoms of B_0 partition Ω ; finite additivity over this partition gives a unique atom a with $\mu(a) = 1$, and $a \neq \emptyset$. Any $\omega \in a$ has $\delta_\omega \upharpoonright B_0 = \mu \upharpoonright B_0 \supseteq s_0$, since every element of B_0 either contains a or misses it. $\square$

Thus a σ -essential contextual state is impossible on a Boolean carrier for a second reason under the amendment: coherence itself already produces the Dirac. The theorem below shows a non-distributive carrier where coherence produces only a **non-Dirac finitely additive** state and rigidity then bars every σ -additive one.

2. The carrier

Lemma 2.1 (Ulam matrix). There is a family $\{C_{\alpha,n} : \alpha < \omega_1, n < \omega\}$ of subsets of M such that:

1. (row disjointness) $C_{\alpha,n} \cap C_{\alpha,m} = \emptyset$ for $n \neq m$;
2. (row covering) $M \setminus \bigcup_{n < \omega} C_{\alpha,n} = \{\beta : \beta \leq \alpha\}$, a countable set;
3. (column disjointness, exact) $C_{\alpha,n} \cap C_{\beta,n} = \emptyset$ for $\alpha \neq \beta$.

Proof. For each $\beta < \omega_1$ fix an injection $g_\beta : \{\gamma : \gamma < \beta\} \rightarrow \omega$ (initial segments are countable; simultaneous choice by AC). Put $C_{\alpha,n} = \{\beta : \alpha < \beta < \omega_1, g_\beta(\alpha) = n\}$. (1): $g_\beta(\alpha)$ is a single value. (2): every $\beta > \alpha$ lies in $C_{\alpha,g_\beta(\alpha)}$. (3): $\beta \in C_{\alpha,n} \cap C_{\alpha',n}$ with $\alpha \neq \alpha'$ would give $g_\beta(\alpha) = n = g_\beta(\alpha')$, contradicting injectivity. $\square$

Definition 2.2 (the carrier L). Let *cells* be $D_{\alpha,n} := C_{\alpha,n} \times F \subseteq \Omega$. Let L be the least σ -class on Ω containing:

- every singleton $\{\omega\}$, $\omega \in \Omega$;
- every cell $D_{\alpha,n}$;
- the cores A, B, C .

Rows ignore the fiber coordinate entirely; cores ignore the M coordinate entirely. This separation is the design principle: the Ulam machinery and the contextual geometry are prevented from interacting except through countable sets.

Lemma 2.3 (σ -class basics). In any σ -class L containing all singletons:

1. (relative complement) $X \subseteq Y$, $X, Y \in L \Rightarrow Y \setminus X \in L$. *Proof:* X and $Y^\perp$ are disjoint, so $X \cup Y^\perp \in L$; its complement is $Y \setminus X$.
2. Every countable $N \subseteq \Omega$ is in L (countable disjoint union of singletons).
3. (countable perturbation) $X \in L$, N countable $\Rightarrow X \cup N, X \setminus N \in L$. *Proof:* $X \cup N = X \sqcup (N \setminus X)$, a disjoint union of members; $X \setminus N = (X^\perp \cup N)^\perp$ with $X^\perp \cup N \in L$ by the previous sentence.
4. Every two-valued state s on L is monotone. *Proof:* $s(Y) = s(X) + s(Y \setminus X) \geq s(X)$ by (1).

3. The trace invariant

Definition 3.1 (the invariant P). For $E \subseteq \Omega$, write $\text{tr}(E) = (E_1, E_2, E_3, E_4)$, the quadruple of fiber traces. Say E satisfies the invariant, written $E \in \widetilde{P}$, if there exist $\xi \subseteq M$ and $\kappa \in E_4$ with

$$E_f \approx \xi^{(\kappa_f)} \quad (f = 1, 2, 3, 4).$$

Call (ξ, κ) a *representation* of E . (Quotient view: in $\bar{Q} = \mathcal{P}(M)/\text{ctble}$, the class quadruple of E equals $\text{diag}([\xi]) \Delta \kappa$. Intrinsically: $E \in \widetilde{P}$ iff all pairwise symmetric differences $E_f \Delta E_g$ are countable or co-countable, and the parity vector they induce is even.)

Lemma 3.2 (representation rigidity). If (ξ, κ) and (η, λ) both represent E with $\kappa, \lambda \in E_4$, then either $\lambda = \kappa$ and $\eta \approx \xi$, or $\lambda = \kappa \Delta 1111$ and $\eta \approx \xi'$. Consequently:

1. each $E \in \widetilde{P}$ has a unique representation with $\kappa \in N$, up to $\approx$ -change of ξ ;
2. E is countable iff its normalized representation is $(\xi, 0000)$ with $\xi \approx \emptyset$;
3. if $\kappa \in N \setminus \{0000\}$ (a *weight-2* element) then E is uncountable, since some trace is $\approx \xi$ and some is $\approx \xi'$, and these cannot both be countable.

Proof. If $\kappa_f = \lambda_f$ for some f , then $\xi^{(\kappa_f)} \approx \eta^{(\kappa_f)}$ gives $\xi \approx \eta$; substituting back into the remaining coordinates forces $\kappa_g = \lambda_g$ for all g (since $\xi \approx \xi'$ is impossible, M being uncountable). Otherwise $\kappa_f \neq \lambda_f$ for all f , i.e. $\lambda = \kappa \Delta 1111$, and any coordinate gives $\eta \approx \xi'$. Since exactly one of $\kappa, \kappa \Delta 1111$ has fourth coordinate 0, (1) follows. (2), (3) are immediate. $\square$

Lemma 3.3 (disjointness table — the central lemma). Let $E, G \in \widetilde{P}$ have normalized representations $(\xi, \kappa), (\eta, \lambda)$, $\kappa, \lambda \in N$, and suppose $E \cap G = \emptyset$ (exact disjointness suffices; only $E_f \cap G_f$ countable for each f is used). Coordinatewise, disjointness mod countable of $\xi^{(\kappa_f)}$ and $\eta^{(\lambda_f)}$ reads:

$$(\kappa_f, \lambda_f) = (0, 0) : \xi \cap \eta \approx \emptyset; \quad (0, 1) : \xi \subseteq \eta \text{ mod ctble}; \quad (1, 0) : \eta \subseteq \xi \text{ mod ctble}; \quad (1, 1) : \xi \cup \eta \approx M.$$

The ten unordered pairs (κ, λ) then resolve as:

Case	(κ, λ)	Forced relations	Conclusion
(I)	$(0000, 0000)$	$\xi \cap \eta \approx \emptyset$	$E \sqcup G \in \widetilde{P}$ with rep $(\xi \cup \eta, 0000)$
(II)	$(\kappa, \kappa), \kappa \text{ weight-2}$	$\xi \cap \eta \approx \emptyset$ and $\xi \cup \eta \approx M$	$\eta \approx \xi'$; $E \sqcup G \approx$ co-countable: rep $(M, 0000)$
(III)	$(0000, \lambda), \lambda \text{ weight-2}$	$\xi \subseteq \eta$ and $\xi \cap \eta \approx \emptyset$ mod ctble	$\xi \approx \emptyset$: E countable ; $E \sqcup G \in \widetilde{P}$ with rep (η, λ)
(IV)	$\kappa \neq \lambda$ both weight-2	$\xi \approx \eta, \xi \cap \eta \approx \emptyset,$ $\xi \cup \eta \approx M$	impossible ($\emptyset \approx M$)

Proof. The coordinate translations are immediate. (I)–(III) are read off. For (IV): any two distinct elements of $N \setminus \{0000\}$ agree in exactly one of the first three coordinates and both vanish

in the fourth, so the multiset of coordinate pairs (κ_f, λ_f) is exactly $\{(1, 1), (1, 0), (0, 1), (0, 0)\}$. Then $(1, 0)$ and $(0, 1)$ give $\xi \approx \eta$; $(0, 0)$ gives $\xi \cap \eta \approx \emptyset$, hence $\xi \approx \emptyset$; $(1, 1)$ gives $\xi \cup \eta \approx M$, hence $\emptyset \approx M$ — false since M is uncountable. $\square$

Row (IV) is the theorem in miniature: no element of L carrying one core pattern is ever exactly disjoint from an element carrying a different core pattern (beyond countable degeneracy). This is what makes the three simultaneous assignments $s_0(A) = s_0(B) = s_0(C) = 1$ immune to additivity constraints: the parity-triangle certificate that killed the subordinate design cannot form.

Lemma 3.4 (family trichotomy). Let $\{E_i\}_{i < \omega} \subseteq \widetilde{P}$ be pairwise exactly disjoint. Then exactly one of:

1. every E_i with uncountable class is diagonal ($\kappa_i = 0000$, $\xi_i \not\approx \emptyset$), pairwise $\xi_i \cap \xi_j \approx \emptyset$;
2. exactly one E_{i_0} is weight-2; all other E_i are countable;
3. exactly two E_{i_0}, E_{i_1} are weight-2, in the **same** coset with $\xi_{i_1} \approx \xi'_{i_0}$; all other E_i are countable.

Proof. By 3.3(IV), all weight-2 members share one coset. By 3.3(II), two same-coset weight-2 members are complementary mod countable; a third in that coset would be $\approx$ -complementary to both, hence the first and third satisfy $\xi \approx \xi'' = \xi$ and $\xi \approx \xi'$ simultaneously — impossible. So there are at most two weight-2 members. If at least one weight-2 member exists, any diagonal member is disjoint from it, hence countable by 3.3(III); if two (complementary) weight-2 members exist, any further member is disjoint from both, and its class is $\leq$ the complement of their join $\approx \emptyset$, hence countable. $\square$

Theorem 3.5 (Normal Form). $L \subseteq \widetilde{P}$: every element of L satisfies the invariant.

Proof. $\widetilde{P}$ contains the generators: singletons and $\emptyset$ are $(\emptyset, 0000)$; Ω is $(M, 0000)$; cells are $(C_{\alpha, n}, 0000)$; A, B, C are $(\emptyset, 1100), (\emptyset, 1010), (\emptyset, 0110)$. $\widetilde{P}$ is complement-closed: $(E^\perp)_f = (E_f)'$, so $(\xi, \kappa\Delta 1111)$ — equivalently $(\xi', \kappa\Delta 1111)$ normalized via 3.2 — represents $E^\perp$. $\widetilde{P}$ is closed under countable pairwise exactly-disjoint unions: apply 3.4 to the family; in case (1) the union is represented by $(\bigcup_i \xi_i^*, 0000)$ where $\xi_i^* := \xi_i \setminus \bigcup_{j < i} (\xi_i \cap \xi_j)$ trims the pairwise-countable overlaps (each $\bigcup_{j < i} (\xi_i \cap \xi_j)$ is countable, so $\xi_i^* \approx \xi_i$, and the ξ_i^* are exactly disjoint, so coordinatewise the union's trace is $\approx \bigcup \xi_i^*$); in case (2) by $(\xi_{i_0}, \kappa_{i_0})$ (the countable members perturb each trace countably); in case (3) by $(M, 0000)$. Hence $\{E : E \in \widetilde{P}\}$ is a σ -class containing the generators, so it contains L . $\square$

4. Structural corollaries

Corollary 4.1 (the missing meet region). No element of L is $\approx$ -equal to $A \cap B = M \times \{1\}$; in particular $A \cap B \notin L$.

Proof. Its trace pattern is $(M, \emptyset, \emptyset, \emptyset)$, class pattern 1000: for a representation, $\xi \in \{\approx \emptyset, \approx M\}$ is forced (traces take at most the two values ξ, ξ' mod countable, and here they take the values $\emptyset$ -class and M -class), giving pattern κ or $\kappa\Delta 1111$ for $\kappa \in E_4$; but 1000 and 0111 have odd weight. $\square$

Corollary 4.2 (stripping bar). If $E \in L$ and $E \subseteq A \bmod \text{countable}$ (i.e. E_3, E_4 countable), then E is countable or $E \approx A$ (traces $\approx (M, M, \emptyset, \emptyset)$). The same holds for B, C and all six core complements.

Proof. Normalized (ξ, κ) : if $\kappa = 0000$, coordinates 3, 4 give $\xi \approx \emptyset$, so E is countable. If $\kappa = 1100$: coordinates 3, 4 are $\approx \xi'$, so $\xi \approx M$ and $E \approx A$. If $\kappa = 1010$: coordinate 3 gives $\xi \approx \emptyset$, coordinate 4 gives $\xi' \approx \emptyset$ — impossible. $\kappa = 0110$: coordinate 3 gives $\xi \approx \emptyset$, coordinate 4 gives $\xi' \approx \emptyset$ — impossible. $\square$

Corollary 4.3 (incompatibility). A and B are incompatible in L ; hence L is non-Boolean.

Proof. Compatibility in a concrete orthomodular poset means $A = A_1 \sqcup D$, $B = B_1 \sqcup D$ with $A_1, B_1, D \in L$ pairwise disjoint; then $D = A \cap B$ (from $A_1 \cap B_1 = \emptyset$, $A \cap B = (A_1 \cap B_1) \sqcup (A_1 \cap D) \sqcup (D \cap B_1) \sqcup D = D$), contradicting 4.1. $\square$

Corollary 4.4 (not a lattice). $A \wedge B$ does not exist in L : the lower bounds of $\{A, B\}$ in L are exactly the countable subsets of $M \times \{1\}$ (by 4.2 applied twice: $E \subseteq A \cap B$ exactly forces, along each core, countability — the case $E \approx A$ is barred by $E \subseteq B$ since $A \not\approx$ any subset of $B \bmod \text{countable}$, its trace 2 being co-...: directly, $E \subseteq A \cap B$ has E_2, E_3, E_4 countable, and the analysis of 4.2 with coordinates 2, 3, 4 countable kills all three weight-2 cosets and forces $\xi \approx \emptyset$), and among the countable subsets of an uncountable... of $M \times \{1\}$ there is no maximum. L is a σ -complete orthomodular **poset**, not a lattice.

Corollary 4.5 (centre). $Z(L) = \{E \in L : E \text{ countable or co-countable}\}$. The quotient of L by the countable ideal has trivial centre, and the state m of §6 vanishes identically on $Z(L) \setminus \{\Omega\text{-class}\}$.

Proof. ($\supseteq$) Countable E is compatible with every $X \in L$: $X = (X \setminus E) \sqcup (X \cap E)$ with both parts in L (2.3), and $E = (E \setminus X) \sqcup (X \cap E)$ likewise. Complements of central elements are central. ($\subseteq$) Let E be central. Compatibility with A puts $E \cap A \in L$ (proof of 4.3), whose trace is $(E_1, E_2, \emptyset, \emptyset) \bmod \text{nothing}$ — coordinates 3, 4 exactly empty, in particular countable; by the case analysis of 4.2, $[E_1] = [E_2] \in \{\emptyset\text{-class}, M\text{-class}\}$. Compatibility with B gives $[E_1] = [E_3]$, with C gives $[E_2] = [E_3]$, each in $\{0, 1\}$. So the first three trace classes are a common constant $t \in \{0, 1\}$. Membership of E itself in $\widetilde{P}$ then forces the fourth class to equal t : pairwise differences of trace classes must be constant, and the parity of the difference vector must be even, so (t, t, t, q) requires $q = t$. Thus E is countable ($t = 0$) or co-countable ($t = 1$). $\square$

5. Rigidity

Theorem 5.1. Every σ -additive two-valued state on L is Dirac.

Proof. Let s be σ -additive and two-valued. If $s(\{\omega\}) = 1$ for some ω , then by monotonicity (2.3.4) $s(E) = 1$ whenever $\omega \in E$, and $\omega \notin E$ gives $s(E^\perp) = 1$, so $s(E) = 0$: $s = \delta_\omega$.

Otherwise s vanishes on all singletons, hence, by σ -additivity, on all countable sets. Fix a row α : the cells $\{D_{\alpha, n}\}_{n < \omega}$ are pairwise exactly disjoint (2.1.1), their union $U_\alpha = (\bigcup_n C_{\alpha, n}) \times F$ lies in L (σ -class closure), and $\Omega \setminus U_\alpha$ is countable (2.1.2), so $s(U_\alpha) = 1$. σ -additivity over the row yields exactly one $n(\alpha) < \omega$ with $s(D_{\alpha, n(\alpha)}) = 1$. The map $\alpha \mapsto n(\alpha)$ sends ω_1 into ω , so some fiber is uncountable: fix $\alpha \neq \beta$ with $n(\alpha) = n(\beta) = n^*$. By column disjointness (2.1.3),

$D_{\alpha,n^*} \cap D_{\beta,n^*} = \emptyset$ **exactly**; their union is in L , and additivity gives $s(D_{\alpha,n^*} \sqcup D_{\beta,n^*}) = 2 -$ contradicting $s \leq 1$. So the second case is void. $\square$

Note what the proof consumes: singletons, cells, rows, columns, and σ -class closure — nothing about cores, and nothing about the invariant. Rigidity is therefore robust to any enlargement of the generator set that preserves the listed items.

6. The state m

Definition 6.1. Fix an ultrafilter $\mathcal{U}$ on M extending the co-countable filter (exists by ZFC; the co-countable filter is proper since M is uncountable). Write $u(\xi) \in \{0, 1\}$ for its characteristic function; u vanishes on countable sets, so u is constant on $\approx$ -classes, $u(\xi') = 1 - u(\xi)$, and $u(\xi \cup \eta) = u(\xi) + u(\eta)$ whenever $\xi \cap \eta \approx \emptyset$ (both cannot hold u -value 1, else $u(\xi \cap \eta) = 1$ for a countable set).

For $E \in L$ with normalized representation (ξ, κ) , define the **vote**

$$v(E) := \kappa \Delta (u(\xi) \cdot 1111) \in E_4,$$

and with $\hat{S} := \{1111, 1100, 1010, 0110\}$ (one member from each complement pair of E_4),

$$m(E) := \begin{cases} 1 & v(E) \in \hat{S}, \\ 0 & \text{else.} \end{cases}$$

Lemma 6.2 (well-definedness). v , hence m , is independent of the representation. Moreover $v(E^\perp) = v(E) \Delta 1111$, so $m(E^\perp) = 1 - m(E)$; $m(\Omega) = 1$; m vanishes on countable sets; $m(A) = m(B) = m(C) = 1$.

Proof. By 3.2 the representation is unique up to $\approx$ -change of ξ (on which u is constant) once $\kappa \in N$; if instead one computes from the non-normalized partner $(\xi', \kappa \Delta 1111)$: $\kappa \Delta 1111 \Delta (u(\xi') \cdot 1111) = \kappa \Delta ((1 + u(\xi')) \cdot 1111) = \kappa \Delta (u(\xi) \cdot 1111)$. Complement: representation $(\xi, \kappa \Delta 1111)$, vote $= v(E) \Delta 1111$; since $\hat{S}$ selects one member per complement pair, m flips. Ω : $(M, 0000)$, $u(M) = 1$, vote $1111 \in \hat{S}$. Countable E : $(\emptyset, 0000)$, vote $0000 \notin \hat{S}$. Cores: $\xi = \emptyset$, $u = 0$, votes $1100, 1010, 0110 \in \hat{S}$. $\square$

Theorem 6.3 (m is a two-valued state on L). For every finite pairwise-disjoint $\{E_i\} \subseteq L$ with union in L (automatic: L is a σ -class), $m(\bigsqcup E_i) = \sum m(E_i)$.

Proof. It suffices to treat pairs (finite families follow by induction, all partial unions lying in L). Let $E \cap G = \emptyset$, $H = E \sqcup G$, and run Lemma 3.3 on normalized representations.

(I) $\kappa = \lambda = 0000$: $m(E) = u(\xi)$, $m(G) = u(\eta)$ (votes $u \cdot 1111$; $0000 \notin \hat{S}$, $1111 \in \hat{S}$), and H has representation $(\xi \cup \eta, 0000)$ up to countable trimming, so $m(H) = u(\xi \cup \eta) = u(\xi) + u(\eta)$ by 6.1.

(II) same weight-2 coset, $\eta \approx \xi'$: votes $v(G) = \kappa \Delta (u(\xi') \cdot 1111) = v(E) \Delta 1111$, so exactly one of $m(E), m(G)$ is 1; and $H \approx \Omega$ has vote 1111 , $m(H) = 1$.

(III) one member countable: $m(\text{countable}) = 0$ and H has the same normalized representation as the uncountable member (countable perturbation), so $m(H) = m(G) + 0$.

(IV) cannot occur. $\square$

Corollary 6.4. $s_0 := m \upharpoonright B$, where B is the $\perp$ -closure $\{\emptyset, \Omega, A, A^\perp, B, B^\perp, C, C^\perp\}$ of the cores, is a two-valued state on B with $s_0(A) = s_0(B) = s_0(C) = 1$, and m witnesses its finite coherence on L (Definition 1.3).

(B is a legitimate finite $\perp$ -closed sub-orthoposet, and s_0 a state on it, independently of m : all twelve cross-intersections among $\{A, A^\perp, B, B^\perp, C, C^\perp\}$ are full fibers, hence nonempty, so the only additivity constraints inside B are the complement pairs.)

Remark 6.5 (the σ -failure of m is self-witnessing). Suppose u selected a cell in every row: $\forall \alpha \exists n(\alpha) : u(C_{\alpha, n(\alpha)}) = 1$. Pigeonholing $\alpha \mapsto n(\alpha)$ as in Theorem 5.1 produces two disjoint sets of u -value 1 — impossible for an ultrafilter. So some row α_0 has $u(C_{\alpha_0, n}) = 0$ for all n : there $m(D_{\alpha_0, n}) = 0$ for every n while $m(\bigsqcup_n D_{\alpha_0, n}) = m(\text{co-countable}) = 1$. Every ultrafilter fails σ -additivity on some row, by the same pigeonhole that drives rigidity. No genericity demands on the matrix are needed anywhere in the construction.

7. Main theorem

Theorem 7.1 (ZFC). Let L, B, s_0 be as above. Then:

1. s_0 is finitely coherent on L (m extends it);
2. no σ -additive two-valued state on L extends s_0 ;
3. hence s_0 is a σ -essential contextual state on L (Definition 1.4), and Ψ (amended [M, Q. 5.1]) holds in ZFC.

Proof. (1) is Corollary 6.4. (2): by Theorem 5.1 any σ -additive extension is some δ_ω ; by [M, Lemma 3.1], $\delta_\omega \upharpoonright B = s_0$ iff $\omega \in K(s_0) = A \cap B \cap C = \emptyset$. $\square$

Proposition 7.2 (Adm checklist for L).

- *Concrete, σ -complete:* by construction (Definition 2.2).
- *Non-Boolean:* Corollary 4.3; the incompatibility of A, B is “uncountably deep” — no countable perturbation repairs it (Corollary 4.1 is mod-countable).
- *Irreducible modulo the countable ideal:* $Z(L)$ is exactly the countable/co-countable sets (Corollary 4.5); the quotient has trivial centre and m annihilates $Z(L)$. Literal irreducibility is unavailable to **any** carrier containing singletons — which the rigidity method requires — so the amendment is forced by the method, not chosen for convenience.
- *Non-segregated:* the contextual pattern lives entirely in the quotient by the countable ideal, on which the centre is trivial; no central Boolean factor carries it. Moreover L is far from a horizontal sum: columns provide exact cross-row orthogonality, and cells are compatible with cores’ countable perturbations only trivially.
- *Non-Polish-representable:* corollary of the witness property via Derr-Williamson ([M, §4]): Polish-representability implies $\Phi(L)$, contradicting 7.1. (Consistently: $|\Omega| = \aleph_1 \leq \mathfrak{c}$; the failure is structural, not cardinal.)
- *Not a lattice:* Corollary 4.4. The theorem resolves the σ -class (orthomodular poset) form; see §9.

Remark 7.3 (an artifact stronger than the witness). In the quotient of L by the countable ideal, Dirac states die (they concentrate on singletons) and non-Dirac σ -additive states are barred by Theorem 5.1. So the quotient is a σ -complete non-Boolean orthomodular poset carrying finitely additive two-valued states (e.g. the induced $\bar{m}$) but **no σ -additive two-valued state whatsoever**. This is starker than the witness itself and may merit separate statement.

Remark 7.4 (where the coherence lives). The finitely additive extension m is built from an ultrafilter on $\mathcal{P}(\omega_1)/\text{ctble}$ — precisely the object [M, §4(3)] observed the carrier cannot *host* as a filter of its own elements. The construction shows the carrier does not need to host it: it borrows only the finite trace of the ultrafilter through the vote map, one Boolean level down. The non-distributivity that blocks the Ulam-style *upper* bound ([M, Rem. 3.7]) is the same non-distributivity that lets the three core assignments coexist (Lemma 3.3(IV)).

8. Verification map and Lean notes

Load-bearing joints, in descending order of residual suspicion.

1. **Lemma 3.4 → Theorem 3.5** (family trichotomy and the countable-union induction). Most quantifier traffic; the trimming construction ξ_i^* and the “third complementary member” exclusion deserve independent derivation. Everything downstream (4.1–4.5) is finite case analysis on top of it.
2. **Lemma 3.3(IV)** — re-derive the coordinate-pair multiset claim for each of the three coset pairs by hand; it is three lines each, and it is the single point the whole theorem balances on.
3. **Lemma 6.2/6.3 well-definedness and case (II)** — the double-representation book-keeping.
4. Theorem 5.1 and Lemma 2.1 are classical (Ulam) and should verify quickly.
5. Proposition 1.6 and Remark 1.5 fix the definitional frame; formalizing 1.6 first is a good warm-up target and guards the amendment.

Lean encoding suggestions.

- **Avoid quotient types:** state the invariant as in Definition 3.1, with $\approx$ as `fun X Y => (X Δ Y).Countable on Set M`. All lemmas are then about `Set.Countable` bookkeeping; Mathlib’s `Set.Countable` API (union, mono, image) suffices.
- **M :** any type with $\sim$ `Countable M` plus a well-order whose proper initial segments are countable; concretely `(Cardinal.aleph 1).ord.toType`, or axiomatize the two properties and instantiate later. Lemma 2.1 needs, for each β , an injection of $\{\gamma // \gamma < \beta\}$ into $\mathbb{N}$ — from countability of initial segments via `Set.Countable.exists_injective` (choice).
- **σ -class:** an inductive predicate `GenSigmaClass (G : Set (Set Ω)) : Set (Set Ω) → Prop` with constructors for generators, empty, univ, compl, and `iUnion` over $\mathbb{N}$ -indexed pairwise-disjoint families. Theorem 3.5 is induction over this predicate; Lemma 2.3 items are two-step derivations inside it.
- **Ultrafilter:** `Ultrafilter.of` applied to the co-countable filter (`Filter.cocountable`, NeBot from uncountability of M). The three properties in 6.1 are `Ultrafilter` API plus “countable sets are not in the co-countable filter’s extension” — the last needs $u \leq \text{cocountable} \rightarrow \forall N \text{ countable}, N \notin u$, immediate from the extension property.

- **Pigeonhole** (5.1, 6.5): “if $\omega_1 = \bigcup (n : \mathbb{N}), S_n$ then some S_n is uncountable”: contrapositive of “countable union of countable is countable” (`Set.countable_iUnion`).
 - E_4 , **votes**: $\text{Fin } 4 \rightarrow \text{Bool}$ with even-parity predicate; the table 3.3 is a decide-able finite case split once the four coordinate translations are lemmas.
 - **Suggested order**: $1.6 \rightarrow 2.1 \rightarrow 2.3 \rightarrow 3.2 \rightarrow 3.3 \rightarrow 3.4 \rightarrow 3.5 \rightarrow 4.1\text{--}4.5 \rightarrow 5.1 \rightarrow 6.1\text{--}6.3 \rightarrow 7.1$.
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9. What remains open

1. **The literal OML form of [M, Q. 5.1]**. Three independent mechanisms now destroy meets exactly where contextual structure enters: loop-generated concrete closures, the even-cardinality logic, and singletons-below-fibers here. **Conjecture**: every concrete σ -complete orthomodular *lattice* satisfies Φ — latticehood forces σ -liftability of finitely coherent patterns. If true, the OMP/OML distinction is precisely the boundary of σ -essential contextuality.
2. **Cardinality**. The witness has $|\Omega| = \aleph_1$. Wright bars finite witnesses; whether a countable Ω can carry one (necessarily without all singletons, or with a different rigidity mechanism) is open, as is any ZFC lower bound beyond “uncountable”.
3. **The quotient artifact** (Remark 7.3): systematic study of σ -classes with $\text{St}_\sigma = \emptyset$ but $\text{St} \neq \emptyset$ — how small, how homogeneous, and whether Polish-representability’s negation admits a combinatorial characterization at this cardinality.